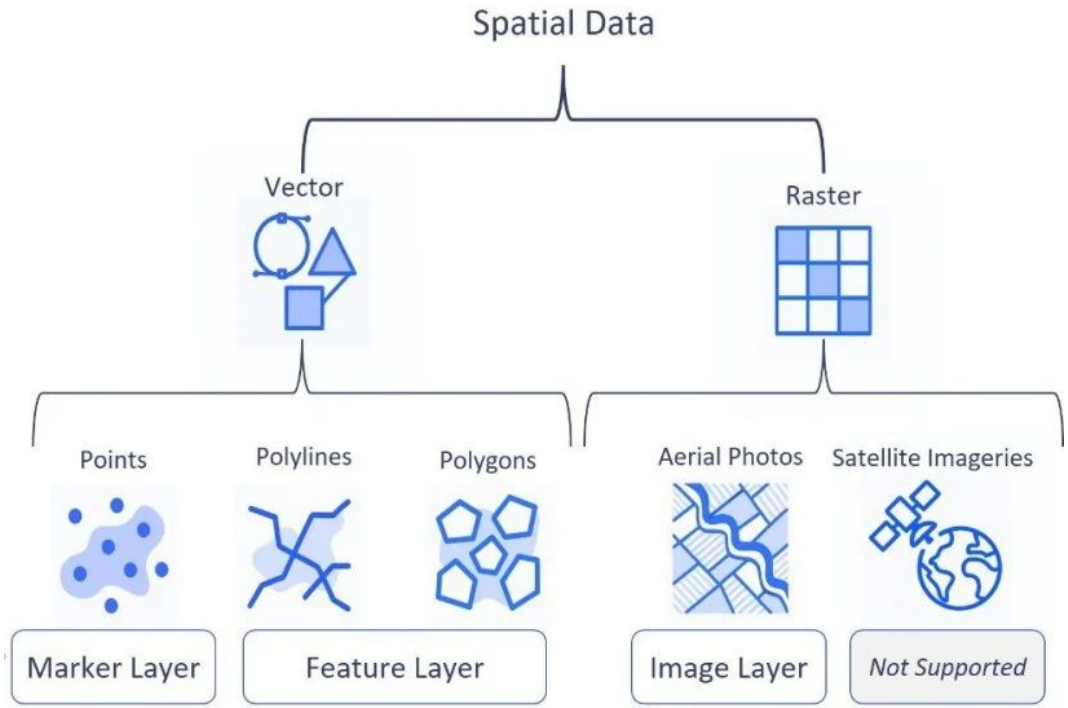
Basics of Remote Sensing Data

FRS 2026. Seminar 4

What is geospatial data?

Geospatial data is time-based data that is related to a specific location on the Earth's surface. It can provide insights into relationships between variables and reveal patterns and trends.

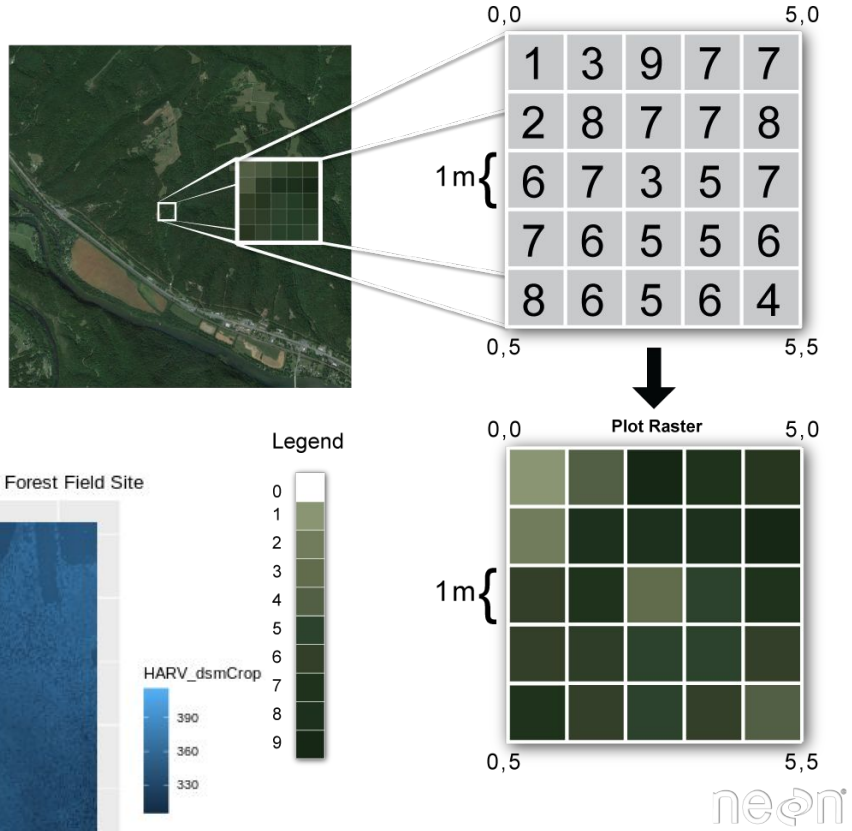
- Geospatial data types:
  - vector and raster
- Components of Geospatial Data
  - Location info
  - Attribute info
  - Temporal info



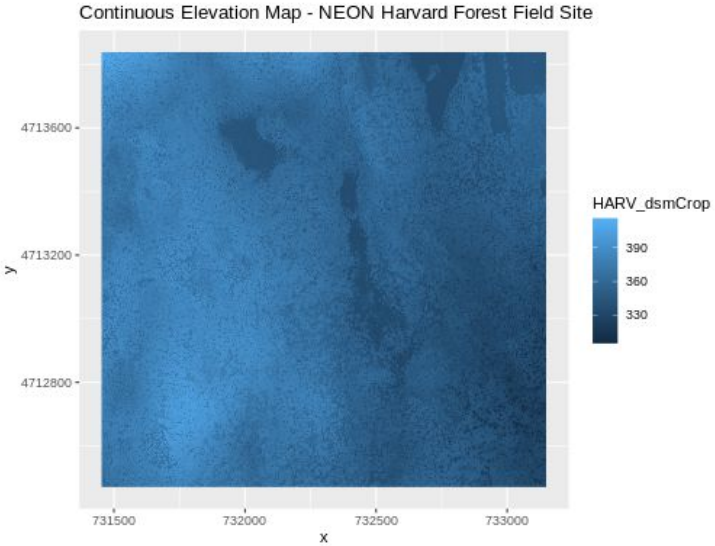
Raster data

Raster data is any pixelated (or gridded) data where each pixel is associated with a specific geographical location. The value of a pixel can be:

- continuous (e.g. elevation)
- categorical (e.g. land use).



A geospatial raster is only different from a digital photo in that it is accompanied by spatial information that connects the data to a particular location.



Raster data. Attributes

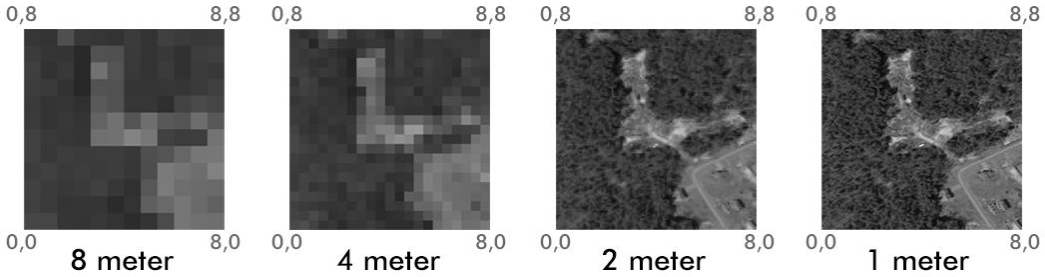
Extent - geographical area covered by a raster (the geographic edge or location that is the furthest north, south, east and west).

Resolution - represents the area on the ground that each pixel of the raster covers.

Other - CRS, Values that represent missing data (`NoDataValue`)

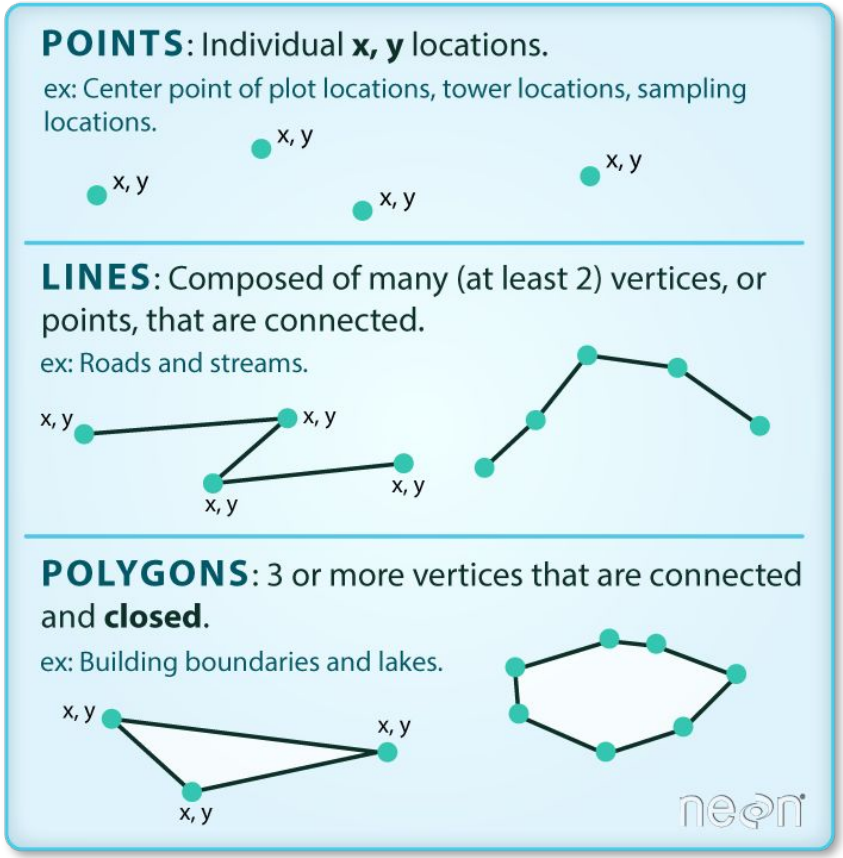
Raster Spatial Extent

Raster over the same extent, at 4 different resolutions



Vector data

Vector data is data in which points, lines and polygons represent physical and abstract features such as properties, cities, roads, mountains and bodies of water.

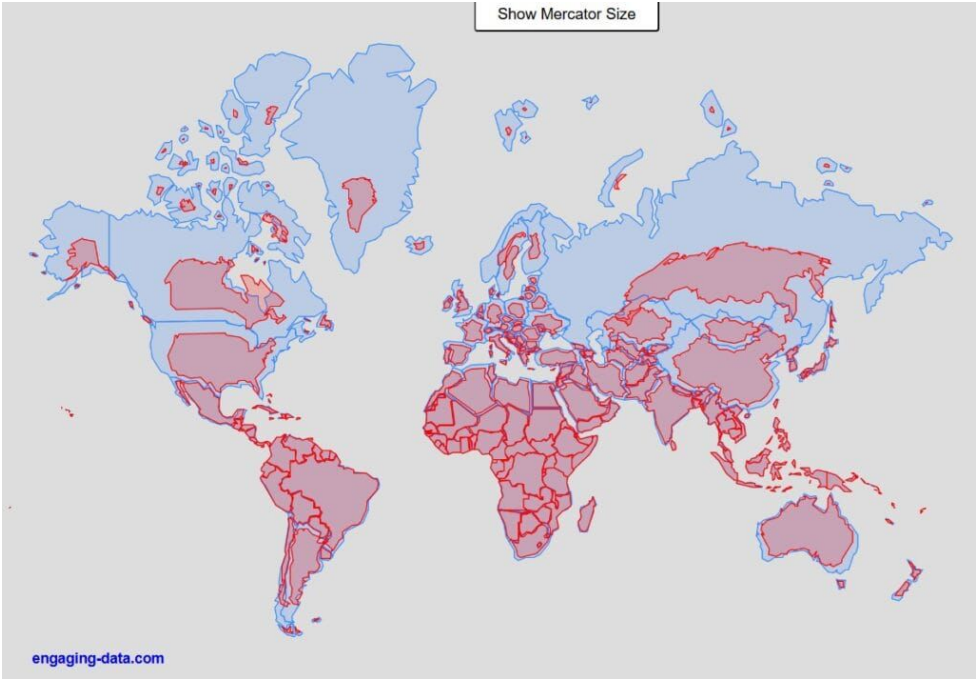


Vector file metadata:

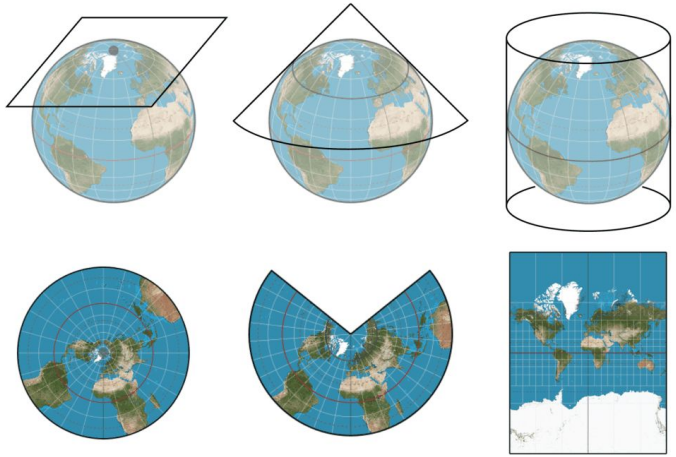
- Extent - the spatial extent of the shapefile (i.e. geographic area that the shapefile covers). The spatial extent for a shapefile represents the combined extent for all spatial objects in the shapefile.
- Object type - whether the shapefile includes points, lines, or polygons.
- Coordinate reference system (CRS)
- Other attributes - for example, a line shapefile that contains the locations of streams, might contain the name of each stream.

Coordinate Reference Systems (CRS)

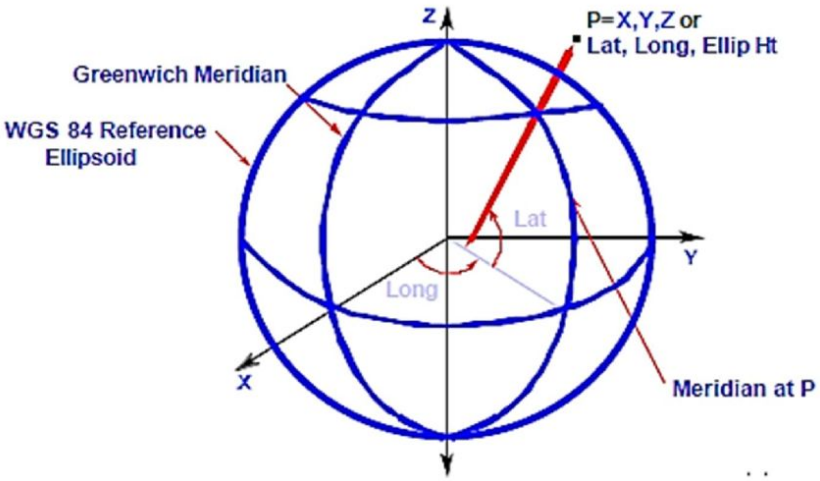
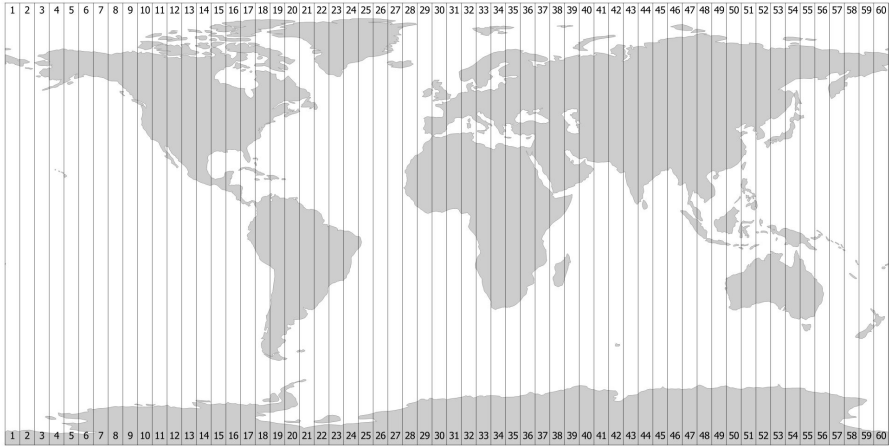
CRS define how geographic data is mapped onto a flat surface. Understanding CRS is essential for ensuring that your data aligns correctly, especially when working with multiple datasets from different sources.



A projection is how you peel your orange and then flatten the peel.



Most common CRS

Most commonly used CRS

- **WGS 84 (EPSG:4326)**
  - Geographic CRS (latitude / longitude)
  - Used by GPS, Google Maps, satellites
  - Units: degrees
  - Best for global datasets and data exchange (also in GPS)
- **Web Mercator (EPSG:3857)**
  - Projected CRS
  - Used in most online maps (Google, OpenStreetMap)
  - Units: meters
  - Good for visualization, but distorts area and distance
- **UTM (Universal Transverse Mercator)**
  - Projected CRS, divided into zones
  - High accuracy for distance and area
  - Units: meters
  - Widely used in local and regional studies